TERUMO BCT AFRICA HACKATHON 2026 · POWERED BY TALO

Technical phase brief

For the 10 shortlisted teams. Everything you need for the 24-hour online build: what to make, the rules, what to submit and how the five finalists will be chosen.
Talo for Terumo BCT · 9 November 2026 · All times are East Africa Time (UTC+3)

- **24 hours**: Tue 10 Nov 12:00 to Wed 11 Nov 12:00
- **10 teams**: build online, from 11 countries
- **5 finalists**: present at the showcase, Fri 13 Nov
- **USD 5,000**: for the overall winner

The challenge

How should a blood-bank information system for Africa work, and how can AI make it better without taking decisions away from people?
In 24 hours you will not build that whole system. You will build **one workflow** of it, well, on realistic data, and show where it fits in the bigger picture. Stay with the challenge area listed for your team in your shortlist email. If your team wants to change, tell us in the kickoff call or on the check-in form before 14:00 on Tuesday.

Challenge area	What a good workflow does
Forecasting	Predicts demand or collections by facility, blood group and week, so managers can plan drives and orders.
Stock-out early warning	Tells a facility it will run out of a blood group before it happens, with how many days are left and why.
Redistribution	Suggests moving units that will expire at one site to a site that is running short, and shows what that saves.
Emergency request and dispatch	Finds the nearest compatible units for an urgent request and gives an honest arrival time.
Cold chain	Spots fridge and cool-box temperature problems, lists the affected units and stops them reaching a patient.
Donors	Brings first-time donors back, plans drives around school holidays and reaches rare blood groups.
Data quality	Finds duplicate, missing and implausible records and helps staff fix them.

Challenge area	What a good workflow does
Analytics	Gives a blood service manager the screen they would open every morning, in plain language.

Key times

When	What happens	Where
Tue 11:45	Call opens. Join with your whole team if you can.	meet.google.com/qao-emcb-gkv
Tue 12:00	**Kickoff and build clock starts.** Welcome, the challenge, the dataset, the rules, questions. 30 minutes.	Same call
Tue 14:00	Send your team roster and your GitHub repository link.	check-in form
Tue 15:00 to 16:00	**Blood-service Q&A.** Ask an expert how blood banks really work. Optional, open to all teams.	meet.google.com/qao-emcb-gkv
Tue 20:00	Check-in 1: two lines on what works and what is blocking you.	check-in form
Wed 06:00	Check-in 2: same two lines.	check-in form
Wed 12:00	**Submissions close.** The form closes on time.	submission form
Wed by 20:00	The five finalists are told by email.	Email
Thu 12 Nov	Finalists: rehearsal and technical check.	Online, times sent with results
Fri 13 Nov	Final showcase in Nairobi and online. Judging and the award.	Details sent to finalists

What to build in 24 hours

Judges look for evidence, not claims: a working screen, a diagram, a test result. Build less and make it work.

Every submission must have

- **A working prototype of one workflow**, running on the starter dataset or other synthetic or openly licensed data. A clickable mock-up alone is not enough.
- **One AI or data feature compared with a simple baseline.** For example, your forecast against last month's average, or your matching against "nearest site first". Show both numbers.
- **A person in control.** Any recommendation that moves blood, contacts a donor or affects a patient needs an Accept, Change or Reject step.
- **A fallback.** The workflow still works if the AI is wrong or offline.
- **An architecture diagram** showing the main parts, the data flows and where your workflow sits in a wider blood system.
- **A README** using the README template, and a **demo video of 3 minutes or less**.

	Qualifies	Competes	Could win
Problem	A general blood-supply problem	A named user and their current workflow	Sized with numbers from the data, checked with the blood-service expert on Tuesday
Prototype	One workflow working on synthetic data	End to end, with two user roles	Plus offline mode or SMS, and an audit trail of who did what
AI and data	A feature that runs	Beats a baseline, with the numbers shown	Shows its reasons in plain language, its uncertainty, and a working fallback
Architecture	A list of modules	A diagram with data flows and APIs	A data model and a mapping to standards such as HL7 FHIR, ISBT 128 or DHIS2
African conditions	Mentioned	Works on a basic phone or a slow connection	Rules, language or branding changed by settings, not code, for a second country

What AI must never do

- Release a unit for transfusion or decide blood group compatibility without a qualified person.
- Override a laboratory result or a quarantine.
- Contact a donor or patient with a message no person has approved.
- Rank patients for blood on anything other than clinical need.

- Present a guess as a fact. Every prediction shows its uncertainty.

If you use a large language model, use it for low-risk jobs such as summaries, drafting donor messages for staff to approve, or translation. Show how you stop it inventing numbers, for example by answering only from your database and showing the query.

The starter dataset

Every team gets the same synthetic year of a blood system, from donor to patient, modelled on Kenya: 47 facilities, 43,000 donors, 49,000 blood units, 17,000 hospital requests, every transfer, daily stock, power cuts and hourly fridge temperatures. It has the problems you applied to solve built in, including shortages during school holidays, slow emergency deliveries, wasted units and messy records.

Download: Applicant's Kit folder on Google Drive. It has the dataset zip and the README template.

The README inside the zip shows which tables to start with for each challenge area and lists the data problems built in on purpose. You may add other data if it is synthetic or openly licensed, and you cite it. **Never use real donor, patient or facility records**, even if someone on your team has access to them.

The rules

Build in the window

All code must be written between Tuesday 12:00 and Wednesday 12:00. Open-source libraries and frameworks are fine. Code written before the hackathon, by you or anyone else, is not. You may plan, read and set up tools before 12:00 on Tuesday.

Show your work

Use one GitHub repository per team, shared with us by 14:00 on Tuesday and public by the deadline. Commit as you go. One large commit at the end will be checked closely.

AI tools are allowed

You may use ChatGPT, Claude, Copilot, Cursor or similar tools. List which ones and what for in your README. Every team member should be able to explain how your solution works. Using AI will not cost you points. Hiding it, or submitting work you cannot explain, will.

Your team

Only the people named on your roster may build. If someone drops out or you want to add someone, tell us before you change anything.

Deadline

The submission form closes at 12:00 on Wednesday. Late submissions are not accepted unless

you reported a technical problem to us before 12:00.

Terms and conduct

Teams keep ownership of what they build. The five finalist teams will receive the participant terms to sign before the showcase. Be respectful in all hackathon channels. Harassment means removal from the hackathon.

What to submit

One person submits for the team, by **Wednesday 11 November, 12:00**, on the submission form. You will need:

- The GitHub repository link, public, with the README filled in
- A demo video of 3 minutes or less (YouTube unlisted, or Google Drive set to anyone with the link)
- A live demo link, if you have one, with test log-ins for the judges
- Up to 8 slides (a link to Google Slides, a PDF or similar)
- Your architecture diagram (in the repository is fine)
- Your baseline and AI result, in a sentence or a small table
- Which AI tools you used and what for
- Who on your team can present in person in Nairobi on Friday, and who would join online

Tip: record your demo video by 10:00 on Wednesday. Everything takes longer near the deadline, and the video is what judges see first.

How the five finalists are chosen

On Wednesday afternoon a panel of judges with technical and blood-service experience scores every submission. Each criterion is scored 1 to 5 and weighted to a total out of 100. The five highest totals go to the showcase. Before scoring, we check every repository against the rules above.

Criterion	Weight	A high score shows
Working prototype	25	The workflow runs end to end on the data, and the demo shows it working, not described.
Technical depth and architecture	20	Sensible design choices, clean data flows, and the team knows where it could fail.
AI and data approach	15	Beats a simple baseline with numbers shown, explains itself, keeps a person in control, and has a fallback.

Criterion	Weight	A high score shows
Blood-system problem and user	15	A specific problem for a named user, using blood-service terms and processes correctly.
Fit for African conditions and scale	15	Works with weak power and connectivity, can be set up for another site or country, and could connect to existing systems.
Demo and readiness to present	10	A clear 3-minute video and a team that is ready to present on Friday.

When totals are within 2 points, the panel looks first at the working prototype score. Because the showcase is in Nairobi and online, the panel may also consider which teams can present in person. The panel's decision is final.

Getting help

We help with

The rules, the dataset, how blood services work, submissions and anything blocking your team that is not a coding problem.

You own

Your technical choices, your code and your debugging. There is no technical help desk, the same for every team.

- **Announcements:** chat.whatsapp.com/GnPADWCUVsWDCfMKrHzlEi. Read-only for teams, so nothing important gets lost.
- **Questions:** the check-in form or email hany@talo.africa. We answer from 08:00 to 23:00. Answers that matter to everyone are posted to all teams, so no team gets an advantage.
- **Something has gone badly wrong** (a team member is ill, you cannot submit): email with "URGENT" in the subject.

If you are a finalist

We will email the five finalists by 20:00 on Wednesday with the showcase programme, your slot, and a time for a rehearsal and technical check on Thursday. The showcase is on Friday 13 November in Nairobi, with teams presenting in person or online. Finalists present the problem and a live demonstration, then take questions from the judges.
The overall winner receives **USD 5,000**, and may have the chance to keep working with Terumo BCT after the hackathon, subject to Terumo's own process.

README template

Copy the README template into README.md at the top of your repository and fill in every section. Judges read it before they watch your video. The template is README_TEMPLATE.md in the Applicant's Kit folder.

Building better blood systems for Africa

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